Experiment # 03

Experimental Proof of OHM's Law OHM's Law in DC & AC Circuit
Linear & Non-Linear Behavior

Learning Objective:

1) In this experiment Ohm's law will be examined, which is a fundamental law in electrical circuits.
2) To learn how to use Ohm's Law in Circuit analysis.

Basic Concepts:

Ohm's law, discovered by a German physicist Simon Ohm (1787-1854), is an important law that defines the relationships between voltage **(V)**, current **(I)**, and resistance **(R)**.
It is often referred to as the foundation of circuit analysis and expressed as,

V = IR

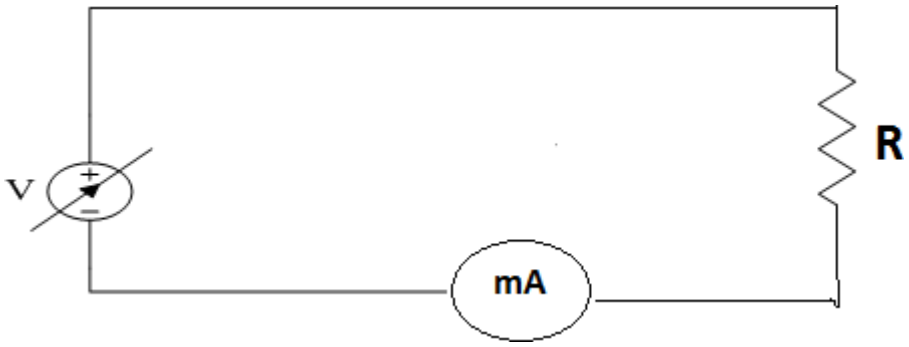
For a given current, an increase in resistance will result in a greater voltage. Alternately, for a given voltage, an increase in resistance will produce a decrease in current.

Equipment

- Adjustable DC Power Supply (0 - 30)
- Digital Millimeter
- Breadboard
- Resistor
- Wires, etc.…

Part-A:

Experimental Circuit:



Procedure

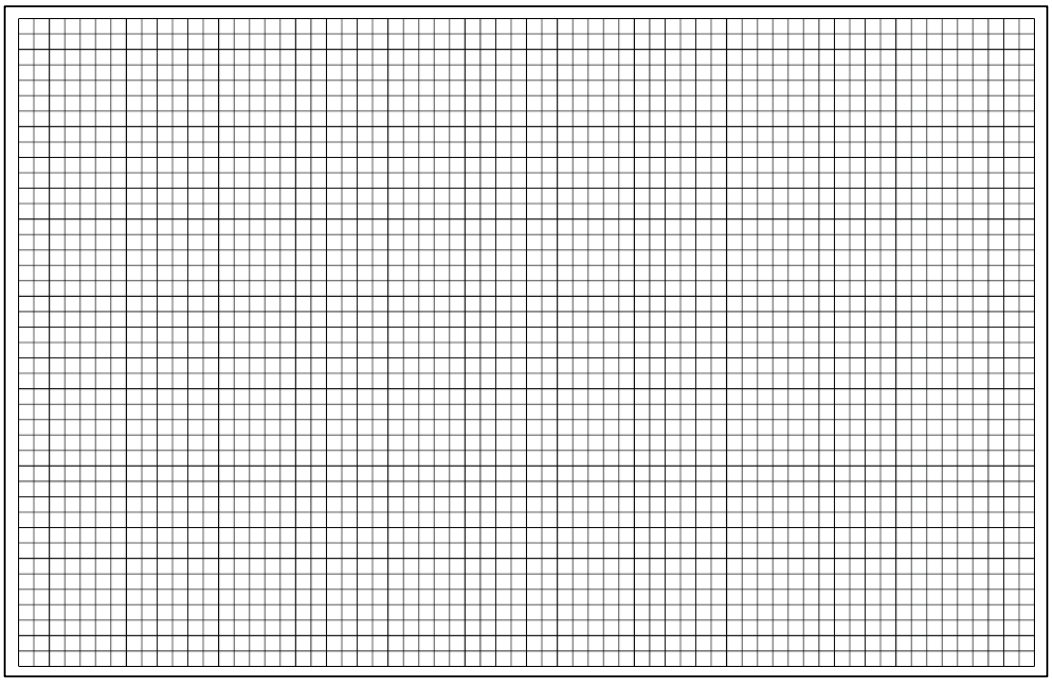
- Build the circuit on Breadboard using the components. Adjust the DMM at DC current to measure DC current and insert it in-line between the source and resistor.
- First calculate the current through the resistor by using above formula and record it in the below Table.
- Now change the power supply voltage from 0 to 30 and record the current reading of the DMM in the table.

- Compare the Measured and Calculated values and if there is any difference then find the % error and write it in % Error column.
- %Error = 100 * (measured value – calculated value) / Calculated Value.
- Now repeat all these steps for another resister values.
- Plot the current versus voltage on the given graph paper. Keep Voltage on the horizontal axis and current on the vertical axis.

Task #1: Fill the given table in accordance with provided procedure.

Sr.#	Voltage	Current Measured	Current Calculated (V=IR)	% Error
1				
2				
3				
4				
5				
6				

Task: Plot Voltage Vs Current. (For both calculated and measured values)



Task #2: Experimental Proof of OHM'S Law with Alternating Voltage source

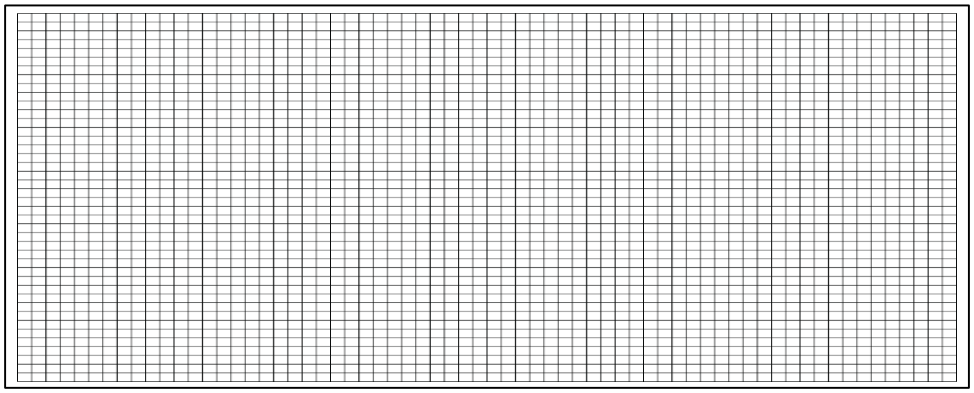
Concept: Instantaneous voltage is the voltage between two points at a particular moment in time, the voltage at a specific instant in time.

RMS voltage is absolutely the most common way to measure/quantify AC voltage. It is also the most useful. Because AC voltage is constantly changing and is at or near the highest and lowest points in the cycle for only a tiny fraction of the cycle, the peak voltage is not a good way to determine how much work can be done by an AC power source. RMS voltage will give you the

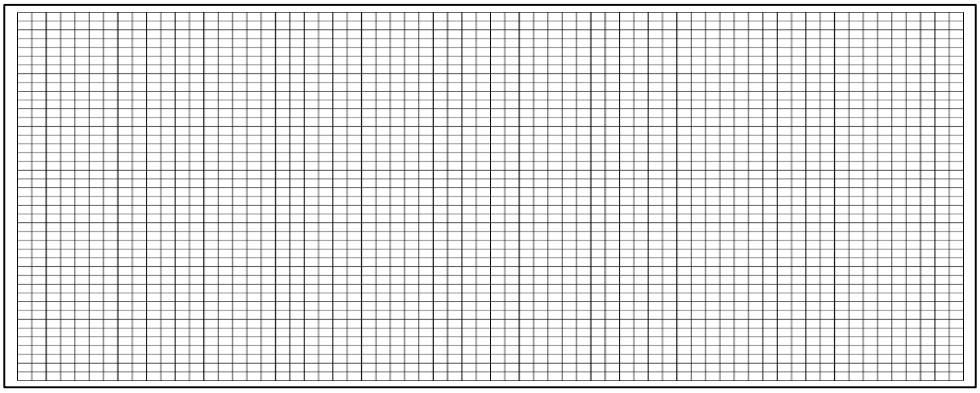
same ability to predict how much work will be done by an AC voltage. The RMS voltage of a *pure*‡ sine wave is approximately 0.707*peak voltage.

Procedure:

1. Use the same circuit as in part A, and replace DC voltage source with an alternating voltage source. Plot Vrms and Irms and see whether you give a straight line as in part A or not.



2. Now plant a **LED** in the circuit used in step 1 and analyze the behavior of circuit. Plot the resistor voltage for this circuit.



TORUBLESHOOTING: -

If the results are not matched, then follow these steps.

1) Make sure that the power supply is on and working properly.
2) Make sure to use components with correct value.
3) Make sure that all grounds are common.
4) Make sure to build circuit properly in accordance with the Figure.
5) Make sure to apply power (+, -) correctly in accordance with circuit Diagram.
6) Check your measuring equipment (Multi-meter, Oscilloscope) is adjusted correctly and you are measuring parameters correctly.
7) If all above steps are correct and you are unable to find the fault, then ask your lab staff for help.

Questions

1. Explain the relation of current flowing through the same resistor when voltage is increased (For DC source).

2. Explain why there is slight difference between calculated and measured current values. (For DC Source)

Conclusion: